

LESSON 13.7

FINDING MISSING ANGLES IN POLYGONS



LESSON INTRODUCTION

8.GR.1.6 Develop and use formulas for the sums of the interior angles of regular polygons by decomposing them into triangles.

LEARNING TARGETS

- I can write and solve algebraic equations to find the measure of an interior angle of a n -gon.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How can I write and solve algebraic equations to find the measure of an interior angle of a n -gon?

LESSON NARRATIVE

This lesson is an extension from prior lessons on angle relationships and writing and solving equations involving triangle angle measures. Students will use their prior experiences to be able to write and solve algebraic equations to find the measure of an interior angle of a n -gon. Students will begin by learning how to find the measures of interior angles using equations. Students will get the opportunity for repeated practice with writing and solving algebraic equations to find missing angle measures in polygons.

COMPONENT SUMMARY

13.7.1 WARM-UP (5 MIN.)

Reflect on career profile video

13.7.2 GUIDED INSTRUCTION (15 MIN.)

Use algebraic equations to find measures of interior angles

13.7.3 GUIDED PRACTICE (15 MIN)

Find measures of interior angles using equations

13.7.4 YOUR TURN! (10 MIN.)

Write and solve algebraic equations to find the measure of an interior angle of a n -gon

13.7.5 WRAP-UP (~5 MIN.)

Reflect on the lesson with a sentence starter activity

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- (Optional) Markers or colored pencils

ADDITIONAL PREPARATION

- 13.7.1 Warm-Up contains a video to accompany the question prompts

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not understand the equation they create to find a missing value; the set-up of the equation should represent the sum of the interior angles for a given n -gon.

THINGS TO CONSIDER

- This lesson is an extension of Lesson 6. Work through each problem ahead of time to look for opportunities to address misconceptions from Lesson 6 and/or to find appropriate places to support students who struggled with Lesson 6.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

13.7.2 Guided Instruction provides students with an introduction to writing an equation for the measure of one interior angle in a polygon. Consider leveraging this routine to assist students with connecting their prior learning to the algebraic methods in this activity. As students share their thinking, make connections to the equation explicit.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

In this lesson, students extend what they learned in Lessons 5 and 6 to solve a more complex set of problems. Scaffolded supports in 13.7.2 Guided Instruction and 13.7.3 Guided Practice are later removed in 13.7.4 Your Turn! as students accurately and efficiently solve questions demonstrating their mastery of the learning target.

DIFFERENTIATE INSTRUCTION

Consider creating an Anchor Chart with visuals, procedures, and examples such as 13.7.2 Guided Instruction question 5. Students can reference the Anchor Chart for guidance with solving similar problems. Consider having students highlight the angles using markers or colored pencils to see the angle they are working with.

13.7.1 WARM-UP: FIGHTING FIRE WITH MATH

TEACHER MOVES

If time permits, review student answers to questions 1 and 2. These are low-stakes questions that provide an opportunity to build interpersonal connections with your students. Understanding how a student perceives a quote or hearing about their experiences gives insight into student thinking and their lives.



13.7.1 Warm-Up: Fighting Fire with Math

Cassie Pennicuik fights fire with math. As a program manager at the County Fire Authority, her job is to lead the growth program. Part of her job is to create a bridge between the technical, strategic, and tactical teams in the organization. She says, "If you really want something in life, go for it and celebrate the effort. In the end, hard work definitely pays off."

1. What do you think she means by "celebrate the effort"?
Answers will vary. Hard work is a positive regardless of the results.
2. Describe a time you went for something that didn't work out, but you still celebrated the effort.
Answers will vary. Students may discuss trying out for a team but not making, learning a skill, or something else they tried.



13.7.2 Guided Instruction: Measures of Interior Angles using Equations

- The measure of an angle in a regular 24-sided polygon is $(12x + 6)^\circ$.

Find the value of each.

Description	Value
Sum of the interior angles	3960°
Measure of one interior angle	165°
x	13.25

- A regular polygon can be decomposed into 43 triangles. The measure of one angle of the polygon is $(17x + 2)^\circ$. Find the value of x .
 $x = 10$
- The measures of two interior angles of a regular polygon are $(9x + 48)^\circ$ and $(15x - 24)^\circ$, respectively.
 - Discuss with your partner how to use the given information to find the value of x .
Set the equations equal to each other and solve for x .
 - Write the equation that can be used to find the value of x .
 $9x + 48 = 15x - 24$
 - Solve the equation.
 $x = 12$
 - What is the angle measure for each interior angle in the polygon?
 156°

13.7.2 GUIDED INSTRUCTION: MEASURES OF INTERIOR ANGLES USING EQUATIONS

TEACHER MOVES

This activity provides students with an introduction to writing an equation for the measure of one interior angle in a polygon. Consider leveraging the compare and connect routine to assist students with connecting their prior learning to the algebraic methods in this activity. As students share their thinking, make connections to the equation explicit.

ENRICHMENT

- How did you set up your equation in question 3b?
- How do you know your equation is correct?

13.7.2 GUIDED INSTRUCTION: MEASURES OF INTERIOR ANGLES USING EQUATIONS (CONTINUED)

TEACHER MOVES

Allow students the opportunity to share out their answers and reasoning to question 4 prior to moving onto question 5. This can provide an opportunity to address any misconceptions before moving forward.

DIFFERENTIATE INSTRUCTION

Consider creating an anchor chart with the procedure and visuals needed to solve question 5. Students can reference this example in the Anchor Chart as they solve similar questions.

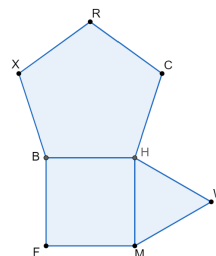
4. Discuss with your partner how to determine the number of sides that a regular polygon has if the sum of the interior angles is given.

- a. Work with your partner to complete the table.

Sum of the Interior Angles	Number of Sides
360°	4
540°	5
720°	6
900°	7

- b. How many sides does a regular polygon have if the sum of the interior angles is $6,300^\circ$?
37

5. Regular polygons $XRCHB$, $BHMF$, and HWM are shown.



The measure of angle WHC is $(19x - 27)^\circ$. Determine the value of x .

$x = 6.789$



13.7.3 Guided Practice: Measures of Interior Angles Using Equations

- How many sides does a regular polygon have if the sum of the interior angles is $10,620^\circ$?
61 sides
- The measures of two interior angles of a regular polygon are $(6x + 18)^\circ$ and $(10x - 86)^\circ$, respectively. What is the angle measure for each interior angle?
Both angles measure 174° .
- A regular polygon can be decomposed into 23 triangles. The measure of one angle of the polygon is $4.5x + 36$. Find the value of x .
 $x = 28.8$

13.7.3 GUIDED PRACTICE: MEASURES OF INTERIOR ANGLES USING EQUATIONS

ENRICHMENT

- What is the relationship between the sum of the angles and the number of sides?
- How can you work backwards to determine the number of sides when you know the interior angle sum?

ENRICHMENT

If a polygon can be decomposed into 23 triangles, how many sides does the polygon have? How do you know?

13.7.4 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Consider using this time to help students who are struggling and may need additional support. If time is limited, consider starting with question 2, since it has multiple parts.

DIFFERENTIATE INSTRUCTION

Consider having students highlight the angles using markers or colored pencils.

ENRICHMENT

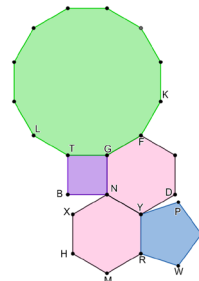
- What information is needed to find the measure of $\angle MRW$?
- What is the measure of each angle in question 2b? How can you use those measures to find the value of m ?



13.7.4 Your Turn!

1. The measure of an angle in a regular 50-sided polygon is $(12.8x - 22.4)^\circ$. Find the value of x .
 $x = 15.25$

2. Five regular polygons are shown.

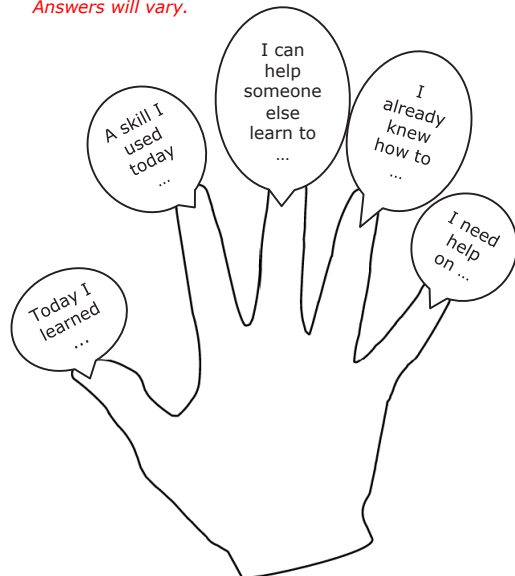


- a. The measure of angle MRW is $(16x - 44)^\circ$. Determine the value of x .
 $x = 11$
- b. The sum of the angles LTG , TBN , NYD , and WRY is $(8m + 52)^\circ$. Find the value of m .
 $m = 52$

**13.7.5 Wrap-Up: Reflection**

1. Reflect on your learning today by completing each sentence. Write your response on each finger.

Answers will vary.

**13.7.5 WRAP-UP: REFLECTION****TEACHER MOVES**

Consider using this formative assessment to decide what to include in a review session before the end of unit assessment.